Definition C.3 (induced path measure)**.** For $m \in \mathbb{N}$, define the *path measure induced by* X^ε on S^m as

$$\widetilde{\mathbb{P}}_x^{\varepsilon,m}(x_{1:m}) := \prod_{i=1}^m p^\varepsilon(x_i|x_{i-1}), \quad x_{1:m} \in S^m, \quad x_0 = x.$$

Similarly to the total variation distance bound between product measures, we have the following result.

Lemma C.4. $\|\widetilde{\mathbb{P}}_x^{\varepsilon,m} - \widetilde{\mathbb{P}}_x^{0,m}\|_{\mathrm{TV}} \le m d_{\mathrm{out}}\varepsilon$.

Proof. Recalling that $\|p^\varepsilon(\cdot|x) - p^0(\cdot|x)\|_{\mathrm{TV}} \le d_{\mathrm{out}}\varepsilon$ for all $x \in S$,

$$\begin{aligned}
&\|\widetilde{\mathbb{P}}_x^{\varepsilon,m} - \widetilde{\mathbb{P}}_x^{0,m}\|_{\mathrm{TV}} \\
&= \frac{1}{2}\sum_{x_{1:m}} \left|\mathbb{P}_x^{\varepsilon,m}(x_{1:m}) - \mathbb{P}_x^{0,m}(x_{1:m})\right| \\
&\le \frac{1}{2}\sum_{x_{1:m}}\sum_{i=1}^m \left|p^\varepsilon(x_i|x_{i-1}) - p^0(x_i|x_{i-1})\right| \prod_{j>i} p^\varepsilon(x_j|x_{j-1}) \prod_{j'<i} p^0(x_{j'}|x_{j'-1}) \\
&= \frac{1}{2}\sum_{i=1}^m\sum_{x_{1:i}} \left|p^\varepsilon(x_i|x_{i-1}) - p^0(x_i|x_{i-1})\right| \prod_{j'<i} p^0(x_{j'}|x_{j'-1}) \\
&\le d_{\mathrm{out}}\varepsilon \cdot \sum_{i=1}^m \sum_{x_{1:i-1}} \prod_{j'<i} p^0(x_{j'}|x_{j'-1}) \\
&= m d_{\mathrm{out}}\varepsilon.
\end{aligned}$$

□

Proposition C.5. *For all $k \in [K]$ and $\varepsilon \le O(M^{-1}(\log M)^{-4})$, it holds that*

$$\sup_{x,y\in C_k}\sup_{z\in S}\left|\mathbb{P}_x(\tau_y^\varepsilon < \tau_z^\varepsilon) - \mathbb{P}_x(\tau_y^0 < \tau_z^0)\right| \le \widetilde{O}\left(\frac{1}{(\log M)^3}\right)$$

and

$$\sup_{x,y\in C_k}\sup_{z\in S}\left|\mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_z^\varepsilon) - \mathbb{P}_x(\bar\tau_y^0 < \bar\tau_z^0)\right| \le \widetilde{O}\left(\frac{1}{(\log M)^3}\right).$$

Proof. Since $\bar\tau_y^\varepsilon < \infty$ for all $\varepsilon \ge 0$ almost surely for $x, y \in C_k$, the above probabilities are well-defined. We prove only the second inequality. Denote the augmented complements $C^{k,z} := C_k^c \cup \{z\}$ and $C^{k,y,z} := C_k^c \cup \{y, z\}$ for brevity. We divide the event $\{\bar\tau_y^\varepsilon < \bar\tau_z^\varepsilon\}$ according to whether the chain has been contained in C_k or has first hit some $w \in C_k^c, w \neq z$ before reaching y, and bound the magnitude of perturbation of each term:

$$\mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_z^\varepsilon) = \mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon) + \sum_{w\in C_k^c\setminus\{z\}} \mathbb{P}_x(X^\varepsilon_{\bar\tau^\varepsilon_{C^{k,y,z}}} = w)\mathbb{P}_w(\bar\tau_y^\varepsilon < \bar\tau_z^\varepsilon). \tag{10}$$

For the first term, we exploit the fast mixing of X^ε within C_k to show a concentration result for $\bar\tau_y^\varepsilon$, then utilize the path measure perturbation bound. Specifically, for m chosen to satisfy (5), the inequality $m \le \bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon$ implies $\tilde X_t^{k,\varepsilon} = X_t^\varepsilon$ for $t < m$, so that

$$\mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon) - \mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon \wedge m) = \mathbb{P}_x(m \le \bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon) \le \mathbb{P}_x(\tilde\tau_y^{k,\varepsilon} \ge m) < \delta \tag{11}$$

Moreover, define $\Gamma_{y,z}$ to be the set of paths γ contained in C_k of length equal to m such that y appears, and first appears before any instance of z, that is

$$\Gamma_{y,z} := \left\{\gamma \in C_k^m : \inf\{k \in [m] : \gamma_k = y\} < (m+1) \wedge \inf\{k \in [m] : \gamma_k \in C^{k,z}\}.\right\}$$

It follows that

$$\mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon \wedge m) = \tilde{\mathbb{P}}_x^{\varepsilon,m}(\Gamma_{y,z})$$

and hence

$$\begin{aligned}
&|\mathbb{P}_x(\bar\tau_y^\varepsilon < \bar\tau_{C^{k,z}}^\varepsilon \wedge m) - \mathbb{P}_x(\bar\tau_y^0 < \bar\tau_{C^{k,z}}^0 \wedge m)| \\
&= |\tilde{\mathbb{P}}_x^{\varepsilon,m}(\Gamma_{y,z}) - \tilde{\mathbb{P}}_x^{0,m}(\Gamma_{y,z})| \le \|\tilde{\mathbb{P}}_x^{\varepsilon,m} - \tilde{\mathbb{P}}_x^{0,m}\|_{\mathrm{TV}} \le m d_{\mathrm{out}} \varepsilon
\end{aligned}$$

by Lemma C.4.

For the second term, by Lemma C.2 we have for all $w \in C_k^c \setminus \{z\}$

$$\mathbb{P}_x(X^\varepsilon_{\bar\tau^\varepsilon_{C^{k,y,z}}} = w) = p^\varepsilon(x,w) + \sum_{u \in C_k \setminus \{y,z\}} \frac{\mathbb{P}_x(\bar\tau_u^\varepsilon < \bar\tau_{C^{k,y,z}}^\varepsilon)}{\mathbb{P}_u(\bar\tau_{C^{k,y,z}}^\varepsilon < \bar\tau_u^\varepsilon)} p^\varepsilon(u,w).$$

The denominator can be lower bounded via a path measure argument similar to before:

$$\begin{aligned}
\mathbb{P}_u(\bar\tau_{C^{k,y,z}}^\varepsilon < \bar\tau_u^\varepsilon) &\ge \mathbb{P}_u(\bar\tau_y^\varepsilon < \bar\tau_u^\varepsilon \wedge m) \\
&\ge \mathbb{P}_u(\bar\tau_y^0 < \bar\tau_u^0 \wedge m) - \|\tilde{\mathbb{P}}_x^{\varepsilon,m} - \tilde{\mathbb{P}}_x^{0,m}\|_{\mathrm{TV}} \\
&\ge \mathbb{P}_u(\bar\tau_y^0 < \bar\tau_u^0) - \mathbb{P}_u(\bar\tau_y^0 \ge m) - \|\tilde{\mathbb{P}}_x^{\varepsilon,m} - \tilde{\mathbb{P}}_x^{0,m}\|_{\mathrm{TV}} \\
&\ge \frac{\nu}{\log M} - \delta - m d_{\mathrm{out}} \varepsilon \ge \frac{\nu}{2\log M}
\end{aligned} \tag{12}$$

by Lemma C.1, as long as $\delta, m\varepsilon = o((\log M)^{-1})$. It follows that

$$\mathbb{P}_x(X^\varepsilon_{\bar\tau^\varepsilon_{C^{k,y,z}}} = w) \le p^\varepsilon(x,w) + \frac{2\log M}{\nu} \sum_{u \in C_k \setminus \{y,z\}} p^\varepsilon(u,w)$$

and

$$\sum_{w \in C_k^c \setminus \{z\}} \mathbb{P}_x(X^\varepsilon_{\bar\tau^\varepsilon_{C^{k,y,z}}} = w) \mathbb{P}_w(\bar\tau_y^\varepsilon < \bar\tau_z^\varepsilon)$$

$$\leq \sum_{w \in C_k^c \setminus \{z\}} p^\varepsilon(x,w) + \frac{2\log M}{\nu} \sum_{u \in C_k \setminus \{y,z\}} \sum_{w \in C_k^c \setminus \{z\}} p^\varepsilon(u,w)$$
$$\leq \left(1 + \frac{2n_{\mathrm{out}} \log M}{\nu}\right) d_{\mathrm{out}} \varepsilon. \tag{13}$$

Now taking $\delta = O(M\varepsilon \log M)$ and $\varepsilon \leq O(M^{-1}(\log M)^{-4})$, we can verify that $\delta = O((\log M)^{-3})$ and

$$m\varepsilon = O\left(M\varepsilon \log \frac{1}{M\varepsilon} \cdot \log M\right) = O\left(\frac{\log\log M}{(\log M)^3}\right).$$

Combining (10), (11) and (13), we conclude:

$$\left|\mathbb{P}_x(\bar{\tau}_y^\varepsilon < \bar{\tau}_z^\varepsilon) - \mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_z^0)\right| \leq m d_{\mathrm{out}} \varepsilon + \delta + O(\log M) \cdot d_{\mathrm{out}} \varepsilon = \widetilde{O}\left(\frac{1}{(\log M)^3}\right),$$

as was to be shown. □

As a corollary, we obtain:

Corollary C.6 (Proposition 2.1 restated)**.** *Any subset $S_\circ = \{x_1, \cdots, x_K\} \subset S$ of cluster representatives $x_k \in C_k$ constitutes a metastable system for X^ε in the sense of* (1) *as $M \to \infty$.*

Proof. For $y \in C_k \setminus \{x_k\}$, it holds that

$$\mathbb{P}_y(\bar{\tau}_{S_\circ}^\varepsilon < \bar{\tau}_y^\varepsilon) \geq \mathbb{P}_y(\bar{\tau}_{x_k}^\varepsilon < \bar{\tau}_y^\varepsilon) \geq \frac{\nu}{2\log M}$$

similarly to (12). On the other hand, for $x_k \in S_\circ$ it follows from Proposition C.5 that

$$\mathbb{P}_{x_k}(\bar{\tau}_{S_\circ \setminus \{x_k\}}^\varepsilon < \bar{\tau}_{x_k}^\varepsilon) \leq \mathbb{P}_{x_k}(\bar{\tau}_{S_\circ \setminus \{x_k\}}^0 < \bar{\tau}_{x_k}^0) + \widetilde{O}\left(\frac{1}{(\log M)^3}\right),$$

and hence (1) follows. □

Now let us study the convergence of the perturbed stationary distributions. Let π^ε for $\varepsilon > 0$ denote the unique stationary distribution of X^ε on S. By the coupling theorem (Meyer, 1989, Theorem 4.1),

$$\pi^\varepsilon = (\xi_1 \pi_1^\varepsilon \ \cdots \ \xi_K \pi_K^\varepsilon) \tag{14}$$

where the coupling factors $\xi_k = \pi^\varepsilon(C_k)$. We then obtain the following corollary of Proposition C.5.

Corollary C.7. *For all $k \in [K]$, it holds that*

$$\sup_{x \in C_k} \left|\frac{\pi_k^\varepsilon(x)}{\mu_k(x)} - 1\right| \leq \widetilde{O}\left(\frac{1}{\log M}\right).$$